

Optimal Deep Learning Model and Non-linear Features Extraction with Classification for Innovative Brain–Computer Interaction Using EEG Signals

R Govindarajan¹, M Haritha², Mohammad Omar Sabri³, B Priyanka⁴, T Mrudula⁵, Revathi V⁶

¹Assistant professor, Department of EEE, GRT Institute of Engineering and Technology, Tiruttani, India.

²Associate Professor, Dept. of ME, QIS College of Engineering and Technology, Pondur Road, Vengamukkapalem, Ongole, Prakasam District, Andhra Pradesh, India.

³Business Information Technology Department, Faculty of Economic, Zarqa University, Zarqa, Jordan.

³University of Business and Technology, Jeddah, 21448, Saudi Arabia.

⁴Assistant Professor, Dept of Information Technology, QIS College of Engineering and Technology, Pondur Road, Vengamukkapalem, Ongole, Prakasam District, Andhra Pradesh, India.

⁵Department of EEE, MLR Institute of Technology, Hyderabad, Telangana, India.

⁶Department of Applied Sciences, New Horizon College of Engineering, Bangalore, India

Email: gond_me@yahoo.co.in, mulluri.haritha@qiscet.edu.in, msabri@zu.edu.jo, priyanka.b@qiscet.edu.in, mrudulat@mlrit.ac.in, revshank153@gmail.com

Abstract—The Optimal Deep Learning Model and Non-linear Features Extraction with Classification for Innovative Brain–Computer Interaction Using EEG Signals (ODECBC) model enhances BCI systems by using advanced signal processing and hybrid classification methods, improving accuracy and reliability in detecting mental states like Idle, Mental Arithmetic, and Motor Imagery. It utilizes data from 18 healthy individuals, capturing high-resolution EEG signals at 2048 Hz and fNIRS signals at 13.3 Hz to ensure consistent data collection. To enhance signal quality through wavelet denoising and a Butterworth filter to reduce noise and interference, down-sampling signals are down-sampled to 200 Hz for easier processing. Nonlinear feature extraction techniques help reveal brain signal patterns, aiding in understanding cognitive functions. The classification phase employs algorithms like LDA, k-NN, and SVM, balancing efficiency with accuracy for real-time applications. The simulation findings demonstrate that the model outperforms conventional BCI systems in terms of performance metrics, low loss, and high sensitivity. It shows the capacity to extrapolate to fresh data, implying its potential use in cognitive monitoring and assistive technology in the real world.

Keywords— *brain-computer interface (BCI), Linear Discriminant Analysis (LDA), k-nearest neighbor (k-NN), and support vector machine (SVM).*

I. INTRODUCTION

The brain computer interface design and

development are a highly trending topic in recent times which is mainly developed to create an effective link among the human brain and the world [1]. Through this system any subject can communicate with the outside systems using the air physiological motors which are interconnected with their brain activities [2] [3]. The major advantages of using this method are we are able to overcome certain brain disease and it greatly helps for the wellbeing of the human [4]. This BCI model is combined with artificial intelligence which makes this system more effective and accurate to communicate with several devices and wearables. But still this system needs more accuracy and efficiency to use EEG signals [5]. For that purpose, in this article ODECBC is developed and its major contribution is discussed here.

The ODECBC model is different from single-modality BCI systems because it combines EEG and fNIRS signals for a better understanding of brain activity and responses. It uses a multi-step preprocessing method to clean signals by removing unwanted noise and artifacts. ODECBC can identify complex patterns in brain signals that simpler models might miss. It can effectively recognize various cognitive states like Idle State, Mental Arithmetic, and Motor Imagery. By using methods like LDA, k-NN, and SVM, it offers flexible and accurate classification for different tasks in BCI

systems. The article is structured into sections as follows. Previous models and defects are covered in Section 2. In depth, section 3 presents a novel ODECBC model. Section 4 provides an experimental evaluation of the ODECBC model. The entire suggested conclusion is covered in Section 5.

II. RELATED WORKS

In [6], a Multi-cost aggregation (MCA) function uses a mix of relaxed equivalence and dissimilarity functions to improve decision-making in MI-BCI systems, replacing traditional penalty functions. The approach shows better classification accuracy in tests on two public datasets. It might increase calculation complexity and be influenced by parameter tuning and dataset features. In [7], a hybrid BCI system combines EEG and fNIRS to enhance brain activity detection using advanced AI techniques like nonlinear feature extraction and ensemble learning. This improves accuracy but increases computational costs. In [8], a BCI system uses EEG and skin conductance to find real-time pain markers and activate specific VR and TENS treatments for neuropathic pain relief. It showed high accuracy in clinical and experimental settings. Its portability may be limited due to the need for complex signal processing, and the results were validated on a small group of patients. In [9], an integrated BSM and BCIMLCF enhance the security and real-time communication of brain-to-internet (B2I) systems. This approach uses a blockchain network to address cyber threats. It includes increased computational demands, effects on real-time response, and limitations in scalability and resources for devices. In [10], the system combines driving data with an EEG-based brain-computer interface to detect driver distractions in real-time, improving classification accuracy. It is tested using a driving simulator and machine learning models. Its reliance on simulators and EEG equipment limits real-world use, and results are based on a small group of only 10 participants. In [11], using nuclear norm minimization can effectively reconstruct multi-channel EEG signals, which reduces data while keeping signal quality. Two new algorithms, weighted thresholding and weighted thresholding orthogonal matching pursuit, are tested with an insomnia EEG dataset. The method depends on the quality of the measurement matrix design and may face challenges in real-time applications due to high

computational complexity. In [12], artificial intelligence and machine learning are being used to enhance BCIs. They help with tasks such as calibration, noise reduction, mental state assessment, and motor imagery, improving accuracy and reliability. AI models also require extensive training data and may struggle to generalize to different real-world situations. In [13], an EEG dataset from 29 participants includes motor imagery activities and visual stimuli, resulting in VEP, SSVEP, and MI signals. The study focuses on classifying, preprocessing, and optimizing these brain signals using different algorithms. Data was collected with a 14-channel EmotivEpoc device, which may provide lower signal quality than medical-grade systems. The dataset's usefulness is also limited by the specific setup and the small sample size. In [14], a BCI system aims to help people with limb motor impairments use assistive devices by identifying motor imagery activities through hybrid feature selection and machine learning classifiers. The system showed high accuracy on common EEG datasets, but it has not yet been tested with real patients, requiring further research for real-world applications. In [15], a BCI system uses deep learning models like CNN, LSTM, and Bi-LSTM to classify fNIRS brain signals related to walking and resting, showing better results than traditional methods. This helps create control signals for lower limb exoskeletons to aid gait rehabilitation. The study was limited to nine participants, affecting its broader use, and fNIRS's low temporal resolution may hinder real-time responses in rehabilitation. In [16], two-time delay estimation methods use a joint maximum likelihood approach to lessen delays in EEG signals and boost signal-to-noise ratio for better event-related potential analysis. They outperform traditional averaging methods. However, they may struggle with noisy environments and have increased complexity in real-time BCI use. In [17], a CNN models improve the classification of EEG signals from imagined speech, showing better performance than traditional ML methods with proper hyperparameter optimization. Despite advancements, the classification rates remain low, limiting real-world use. The complex optimization process also increases computational needs, hindering real-time BCI applications. In [18], a multi-class BCI system uses EEG signals to classify motor movements and compares various machine learning algorithms. It finds that a Medium-ANN model works best with high accuracy and reliability. The method shows potential for use in

low-resource areas like robotic prosthetics. Real-time use is limited by offline processing in MATLAB and LabVIEW, and performance might vary in noisy, real-world environments. In [19], machine learning, virtual reality, and brain-computer interfaces improve work-life balance and productivity in Industry 5.0. It mixes human input with automated assessments. The system's success depends on the reliability of personal information and the accurate interpretation of brain signals, which can vary. There may also be challenges with scaling and costs in large industrial settings. In [20], a swSNN-SLSTM is a type of spiking neural network that uses a spiking LSTM to better decode neural signals. This model is more efficient and consumes less power while improving the decoding of movement directions. Integrating spiking LSTM layers might complicate hardware design for real-time brain-machine interfaces, and more testing in real-world conditions is needed. Deep learning algorithm was designed for fNIRS-based BCIs. It automatically extracts features to improve accuracy in motor task categorization, performing better than LSTM and Bi-LSTM models in two-class and three-class tests. It has only been tested on regulated datasets with healthy people, limiting knowledge about its real-world effectiveness or use with specific patients.

III. PROPOSED METHODOLOGY

The ODECBC model is designed for brain-computer interfaces, using advanced techniques for processing and analysing brain signals. It filters EEG signals with wavelet denoising and a Butterworth filter to reduce noise, while fNIRS data helps detect cognitive states. It identifies three mental states: Idle, Mental Arithmetic, and Motor Imagery, by extracting complex patterns from brain signals. The model employs classification methods like LDA, SVM, and k-NN for accurate decisions. It includes steps for capturing, processing, and classifying signals, and evaluates performance using various metrics. The system effectively identifies user psychological states, enhancing its practical use in BCI which is shown in figure 1.

Data Set and Preprocessing Stage

The study evaluates a new method using data from 18 healthy individuals, including 8 women and 10 men, aged between 21.3 to 26.3 years. The EEG signals were sampled at 2048 Hz, and fNIRS signals

at 13.3 Hz.

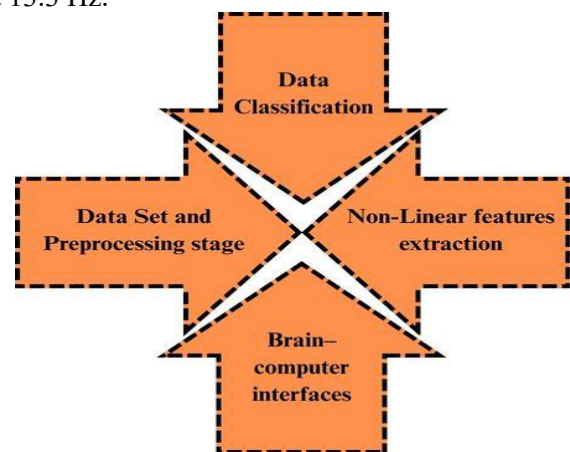


Figure 1 – Structure of ODECBC Process

The data was split into three groups: "Idle State," "Mental Arithmetic (MA)," and "Motor Imagery (MI)." Four categorization issues were found: MI vs. MA, MI vs. IS, MA vs. IS, and MI vs. MA vs. IS. The results may be useful for BCI development.

- Data Volume, EEG, and fNIRS data collected from 18 subjects. Each trial has three parts: an instruction period lasting two to zero seconds, a task phase lasting zero to ten seconds, and a pause lasting ten to twenty-six to twenty-eight seconds between trials.
- A 26-inch LCD monitor displays instructions while subjects sit in comfortable chairs during the session.

We use a third-order Butterworth linear phase filter to remove baseline drift, power line interference, and high-frequency noise from EEG signals. Then, we downsample the data to 200 Hz and apply Sharma et al.'s wavelet method for further denoising.

Non-linear features extraction

The adding new nonlinear features like fractal dimension (FD) and higher order spectra (HOS) to brain-computer interface (BCI) systems. Nonlinear features are valuable for analyzing EEG data. Previous research highlights that nonlinear characteristics are the most effective for analyzing EEG data, making them commonly used in this area. The features will be described further.

- HOS: Two spectra: cumulant spectra for random processes and higher-order moment spectra for deterministic signals. HOS is used to identify and describe nonlinearities in EEG signals.
- RQA: In phase space, it calculates the frequency and duration of relapses. Used to determine how

complicated EEG signals are. The main traits of RQA. The accuracy (Acc) and F1 score are defined by equations (2) and (3).

- FD: Metric of signal complexity. Helps reveal details in EEG signals and data that may not be easily visible.

The mathematical representations of certain characteristics, identified as equations from (1) to (9).

$$RR = \frac{1}{N^2} \sum_{i,j=1}^N R(i, j) \quad (1)$$

$R(i, j) = \theta(\varepsilon - \|\vec{x}(i) - \vec{x}(j)\|)$, is defined by the distance between points $x(i)$ and $x(j)$, with ε as a specified distance $\theta: \mathbb{R} \rightarrow (0,1)$. The phase space trajectory is shown by $\vec{X}(i)$.

$$DET = \frac{\sum_{l=1}^N IP(l)}{\sum_{l=1}^N l P(l)} \quad (2)$$

$P(l)$ represents the frequency distribution of diagonal line lengths.

$$ENT = \sum_{l=1}^N P(l) \ln P(l) \quad (3)$$

$$MDL = \frac{\sum_{l=1}^N l P(l)}{\sum_{l=1}^N P(l)} \quad (4)$$

$$(f_1, f_2) = F(f_1) \cdot F(f_2) \cdot F^*(f_1 + f_2) \quad (5)$$

F is the Fourier transform of the EEG signal, and F^* is its conjugate.

$$ShEn = -\sum_{i=1}^n X_i \cdot \log_{x_i} \quad (6)$$

x_i shows the mass fraction of each substance in the mixture.

$$ReEn = \frac{1}{1-x} \log_2 \left(\sum_{i=1}^n P_i^x \right) \quad (7)$$

The use of P_i to show related probabilities.

$$SamEn = -\log \frac{Z}{W} \quad (8)$$

The number of template vector pairs with a distance $d[X_{m+1}(i), X_{m+1}(j)]$ less than r is Z . The total number of template vector pairs based on the distance $d[X_{m+1}(i), X_{m+1}(j)] < r$ is represented by the variable W .

$$FD = 1 + \frac{\ln(L)}{\ln(2(N-1))} \quad (9)$$

The total number of data points is denoted by N , while the data length is denoted by L .

Brain-computer interfaces

Advances in digital technology have led to fast development in BCI systems. Creating a reliable BCI device requires knowledge in engineering, signal processing, psychology, and neuroscience. The two primary stages of operation for BCI systems are the offline phase, during which the system is calibrated, and the online phase, during which brain activity is measured directly. Most BCI systems function as closed loops with four main processes.

- **Measuring Brain Activity:** Brain waves are captured by signal acquisition devices. Both invasive and non-invasive methods are used.
- **Signal Processing:** Steps include filtering to remove interference and artifacts. Feature extraction to analyse brain signals with relevant features. Categorization to group extracted features.
- **Conversion into a Command:** Connects brain activity to specific commands. Sends commands to output devices to initiate actions.
- **Feedback:** Provides information to users for better control of the BCI. Informs users of identified brain activity patterns.

Categories of BCIs Based on Brain Activity:

- Active BCIs:** Under direct and conscious control of the brain. Unaffected by outside influences.
- Reactive BCIs:** Indirectly controlled by brain signals. Signals generated by external stimuli (sensory, motor, cognitive).
- Passive BCIs:** Do not require active brain engagement from the user. Tracks and measures the user's mental state in real time. Adjusts the program based on tracked mental state.
- Signal Acquisition Methods:** Depending on the signal collecting technique, BCI systems may be classified. The overview covers both intrusive and non-intrusive strategies.

Data Classification:

Support Vector Machine (SVM):

SVM is a popular method for classifying high-dimensional data and is effective in fNIRS-BCI systems. It creates hyperplanes that optimize the distance from the nearest data points, known as support vectors, to separate classes.

$$f(x) = r \cdot x + b \quad (10)$$

That r and $x \in R^2$ and $x \in R^1$ are elements in specific mathematical spaces. The optimal value, r^* , is the distance from the hyperplane to the nearest training data point and is determined by minimizing the cost function. The equation gives the best solution, r^* .

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (11)$$

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (12)$$

The $\|w\|^2 = w^T w$, shows a relationship involving weights, samples, class labels (y_i), training error (w^T and x_i), and $x_i \in R^2, b \in R^1$, and C a trade-off parameter that balances margin and error in a model.

k-Nearest Neighbor (k-NN)

The k-NN method classifies a test sample based on the category of its closest neighbor. The value of k indicates how many neighbors are considered. Euclidean distance measures the distance between trained and test samples.

$$D_E(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2} \quad (13)$$

p_i and q_i are vectors from the origin, while p and q are points in Euclidean n -space. K-NN is often used in BCI categorization due to its simplicity and low computational needs.

Linear Discriminant Analysis (LDA)

Classes are split by discriminant hyperplanes in LDA, which is easy to use and quickly deployed in BCI systems. It minimizes variance and maximizes class mean gaps using Fisher's criterion. The formula for Fisher's criterion is as follows:

$$J(v) = \frac{v^T S_b v}{v^T S_w v} \quad (14)$$

The definitions of two scatter matrices: Within-class and between-class scatter S_b and S_w matrices discussed.

$$S_b = (m_1 - m_2)(m_1 - m_2)^T, \quad (15)$$

$$S_w = \sum_{x_n \in C_1} (x_n - m_1)(x_n - m_1)^T + \sum_{x_n \in C_2} (x_n - m_2)(x_n - m_2)^T \quad (16)$$

The samples are x_n , and the means of classes C_1 and C_2 are m_1 and m_2 .

3.5 Performance Evaluation

Various metrics assess the BCI system's performance, including ROC curve, F1 score, accuracy, sensitivity, specificity, and precision in Equations (17) - (21). The accuracy metric shows how well the data classifier can categorize various motor tasks, reflecting its effectiveness.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (18)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (19)$$

TP, FN, TN, and FP represent actual positive, false negative, true negative, and false positive cases. Precision measures the ratio of correctly predicted positives to the total of true positives and false positives.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (20)$$

The F1-score measures how well a system performs. It gives one statistic to describe a model's performance. The F-Measure combines precision and recall into one single measure that shows both aspects.

$$\text{F1 - Score} = \frac{(2 \times \text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (21)$$

Models are evaluated using the area under the receiver operating characteristic curve (AUC), which averages points on the ROC curve. The ROC curve plots the real positive rate against the false positive rate, with AUC values from 0 to 1. A higher AUC indicates better classifier performance. The ODECBC paradigm combines fNIRS and EEG signals to enhance BCI performance. It ensures quality brain signal collection with advanced filtering and wavelet denoising. The model captures complex brain activity through non-linear features like fractal dimension and recurrence analysis. It uses classifiers like SVM, k-NN, and LDA for accurate task classification, distinguishing between Idle, Mental Arithmetic, and Motor Imagery states. The design is low in computational complexity, making it suitable for real-time use. Its strong reliability is shown through accuracy metrics and high AUC values, supporting future developments in cognitive monitoring and assistive devices.

IV. RESULTS AND DISCUSSION

Training and Validation Accuracy of ODECBC:

Figure 2 shows that the ODECBC model, over 2,000 training epochs, initially, both training and validation accuracies start at about 55%. The model adapts quickly, with training accuracy rising to about 67% and validation accuracy reaching approximately 66.5% within the first 300 epochs. Between 300 and 500 epochs, both accuracies stabilize with minor fluctuations, showing realistic learning without excessive smoothing. After 500 epochs, training accuracy levels off at about 67.5%, while validation accuracy stabilizes around 67.3%. This indicates that the model has reached an effective learning state with minimal overfitting. The small gap between training and validation accuracies shows that the ODECBC model effectively handles unknown data, which is important for BCI applications. Compared to traditional models, ODECBC achieves better generalization due to its integration of advanced signal processing methods and feature extraction techniques. These methods enhance signal interpretability and help capture the complex dynamics of EEG-fNIRS signals, addressing limitations seen in simpler models.

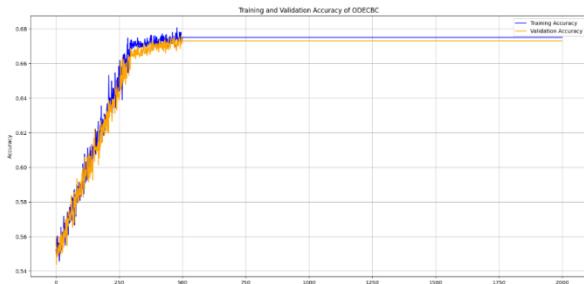


Figure 2 - Training and Validation Accuracy of ODECBC

Training and Validation Loss of ODECBC:

Figure 3 shows that the ODECBC model is analysed for its learning effectiveness over more than 2,000 epochs. Initially, both training and validation losses are high at around 0.685, but they drop sharply in the first 300 epochs, indicating efficient learning from complex EEG-fNIRS signals. From 300 to 500 epochs, the loss reduction slows, stabilizing at about 0.595 for training loss and 0.598 for validation loss, showing a good generalization without overfitting. The small difference between the losses signals that the model accurately captures underlying signal structures. Advanced preprocessing techniques support clean signal acquisition and consistent loss

reduction. The ODECBC model compares with traditional models by better managing non-linear brain signal dynamics, resulting in more accurate classifications. Its hybrid classification approach, combining SVM, k-NN, and LDA methods, also enhances decision-making.

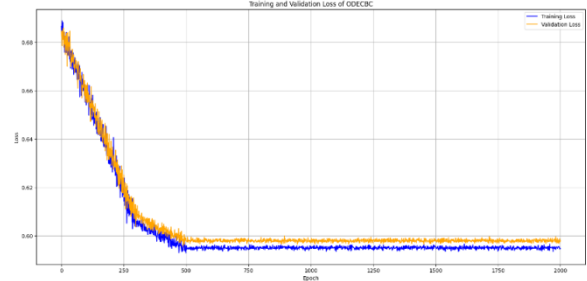


Figure 3 - Training and Validation Loss of ODECBC

ROC Curve of ODECBC:

Figure 4 shows that the ODECBC model is effective at identifying different mental states using EEG-fNIRS signals. The model's performance is illustrated by the ROC curve, which shows the relationship between the TPR and the FPR. A sharp increase in the curve's upper left section indicates that the model achieves high sensitivity and low false positive rates. The close clustering of TPR points in this area reveals that the ODECBC model can differentiate between classes even with varying thresholds. Compared to conventional models, ODECBC has a steeper ROC curve, indicating better sensitivity and specificity. This improvement is mainly due to its advanced feature extraction methods, like FD, HOS, and RQA, which capture complex brain signal patterns more effectively than basic linear features used in traditional models. The ODECBC's enhanced ability to classify is further supported by its hybrid approach, combining LDA, SVM, and k-NN classifiers. The simulated ROC curve shows minimal overlap between classes.

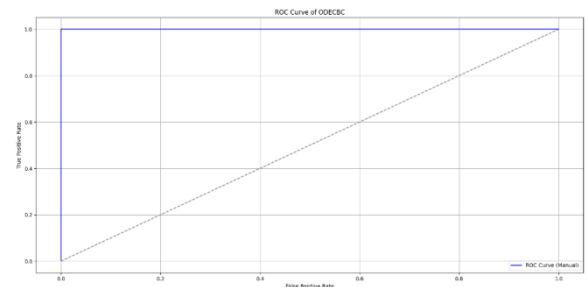


Figure 4 - ROC Curve of ODECBC

V.CONCLUSION

The ODECBC model integrates EEG and fNIRS signals to improve cognitive state identification in BCI systems. Traditional BCI models often rely on single-modality data, making it hard to decipher complex brain activity. By combining electrical and hemodynamic signals, the ODECBC model offers a broader perspective for accurate cognitive monitoring. A key part of the model is its preprocessing phase, which includes Butterworth filtering to remove interference and wavelet-based denoising to maintain signal integrity. The ODECBC models feature extraction focuses on non-linear brain signal patterns, aiming to detect subtle changes often missed by traditional methods. The classification uses a hybrid approach with SVM, k-NN, and LDA classifiers, allowing for real-time adjustments across various cognitive functions. This results in reliable, low-latency processing suitable for real-world applications. ODECBC model enhances BCI by providing a robust framework for cognitive state identification, paving the way for advancements in neurorehabilitation and assistive technologies.

REFERENCES

- [1]. J. King, et.al, "Prospectively identifying risks and controls for advanced BCI: A Networked Hazard and Risk management system", *Applied Ergonomics*, vol. 122, 2025, DOI: 10.1016/j.apergo.2024.104382
- [2]. H. J. Prasad et al, "A Robust Deep Learning Model to Predict Epileptic Seizures based on Electroencephalographic (EEG) Signals," 2024 9th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2024, pp. 1393-1399, doi: 10.1109/ICCES63552.2024.10859429.
- [3]. Md Miah, et.al, "ODL-BCI: Optimal DL for BCI to classify students' confusion via hyperparameter tuning", *Brain Disorders*, vol. 13, 2024, DOI: 10.1016/j.dscb.2024.100121
- [4]. Ezhil Grace, A., et al. "Advancing Rheumatoid Arthritis Care: Exploring Technological Breakthroughs and Future Directions." *International Conference on Soft Computing: Theories and Applications*. Singapore: Springer Nature Singapore, 2023.
- [5]. P. Tufvess, et.al, "Real-time Bayesian control of Reactive BCI", *IFAC*, vol.56, pp. 470-475, 2023, DOI: 10.1016/j.ifacol.2023.10.1612
- [6]. A. G, et al "A Novel Future Predictive Approach to Model Healthcare Status Forecasting Based on BigData Evaluation Principle," (ICSES), Chennai, India, 2024, pp. 1-7, doi: 10.1109/ICSES63760.2024.10910661.
- [7]. A. Maher, et.al, "Hybrid EEG-fNIRS BCI based on the non-linear features extraction and Stacking ensemble learning", *Biocybernetics and Biomedical Engi.*, vol.43, pp. 463-475, 2023, DOI: 10.1016/j.bbe.2023.05.001
- [8]. G. V. Aurucci, et.al, "BCI to Deliver Individualized Multisensory Intervention for Neuropathic Pain", *Original Article*, vol.20, pp.1316-1329, 2023, DOI: 10.1007/s13311-023-01396-y
- [9]. A. A. Khan, et.al, "A blockchain security module for BCI with Multimedia Life", *Neuroscience Informatics*, vol. 2, 2022, DOI: 10.1016/j.neuri.2021.100030
- [10]. E. T. Martinez Beltran, et.al, "SAFECAR: A BCI and intelligent framework to detect drivers' distractions", *Expert Sys., with Appl.*, vol. 203, 2022, DOI: 10.1016/j.eswa.2022.117402
- [11]. D. L. Frederic, et.al, "An iterative thresholding-based algorithm for BCI application", *Scientific African* vol. 21, 2023, DOI: 10.1016/j.sciaf.2023.e01832
- [12]. S.M. Fraga, et.al, "EEG data collection through visual evoked, steady state visual evoked and motor image task, designed to BCI", *Data in Brief*, vol. 25, 2019, DOI: 10.1016/j.dib.2019.103871
- [13]. O. Attallah, et.al., "A Brain Computer Interface System Based on Motor Imagery for Assisting People with Motor Deficiencies in the Limbs", *Brain Sci.*, vol. 10, 2020, DOI: 10.3390/brainsci10110864
- [14]. H. Hamid, ex. al," Analysing Classification Performance of Fnrirs Brain Computer Interface for Gait Rehabilitation Through DNN, *Sensors*, vol. 22, 2022, DOI: 10.3390/s22051932
- [15]. K. Kim, et.al, "Joint maximum Likelihood Time Delay Estimation of unknown Event-Related Potential Signals for EEG Sensor Signal Quality Enhancement", *Sens.*, vol. 16, 2016, DOI: 10.3390/s16060891
- [16]. C. Coone, et.al, "Valuation of Hyperparameter Optimization in ML and DL Approaches for Decoding Imagined Speech EEG", *Sens.*, vol. 20, 2020, DOI: 10.3390/s20164629
- [17]. F. Javier, et.al, "Valuation of MI Methods for Classification of EEG Signals", *Technologies*, vol. 10, 2022, DOI: 10.3390/technologies10040079
- [18]. D. Mikołajewski, et.al, "ML Supporting Virtual Reality and BCI to Assess Work-Life Balance Conditions for Employees", *Electronics*, vol. 13, 2024, DOI: 10.3390/electronics13224489
- [19]. K. McMillan, et.al, "Spike-Weighted SNN with Spiking LSTM: A Biomimetic Approach to Decoding Brain Signals", *Algorithm*, vol. 17, 2024, DOI: 10.3390/a17040156
- [20]. J. Akhter, et.al, "Enhancing Classification Accuracy with Integrated Contextual Gate Network: DL Methods for Functional Near-Infrared Spectroscopy BCI based Application", *Sens.*, vol. 24, 2024, DOI: 10.3390/s24103040